



(AUTONOMOUS)

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Shamshabad, Hyderabad – 501218, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)
(A.Y. 2026-2027)

Major Project – Phase I Submission Form: Review - 1

Date Review: 31-07-2026

1. Team Number: 32
2. Hall Ticket Numbers and Section: 23881A6671, 23881A66A3, 23881A66B8 CSM-B
3. Name of the Supervisor: Mr. N.S.S.S.Girish Kumar
4. Major Project Title: Automated Curriculum Design Using LLMs
5. Technology Used:
 - Frontend: HTML, CSS, JavaScript (or React, if a framework is used).
 - Backend: Python with Flask or FastAPI.
 - Database: SQLite / MySQL / PostgreSQL.
 - AI/ML Stack: Python, scikit-learn, PyTorch/TensorFlow (for Model 1 and Model 3), Hugging Face / LLM API (for Model 2).
 - LLM Integration: OpenAI GPT / Google Gemini / open-source LLM (e.g., Llama, Mistral) via API.
 - Deployment: Local server or cloud (e.g., Render, Railway, AWS).
 - Tools: Git, GitHub, Jupyter/Colab, VS Code.
6. Problem Statement:
 - Teachers spend significant time manually designing curricula, creating assessments, and adapting lessons for different learners. Although LLMs can assist, most existing tools are fragmented and do not provide an end-to-end, teacher-guided workflow. The problem is to develop a system that can automatically generate curricula, create assessments, align content with industry skill demand, adapt to student performance, and still allow teacher review and control.

7. Existing DL Models Applied to the Problem Statement:

- Skill extraction from job descriptions: Models like SKORE, Skill-LLM, SkillNer use NLP/LLMs to extract and classify skills from job postings.
- Knowledge tracing / student proficiency: Deep learning models such as DKT (Deep Knowledge Tracing), DKVMN, and transformer-based knowledge tracing models estimate learner mastery from sequential performance data.
- LLM-based lesson/curriculum generation: Systems like EduPlanner, LessonPlanLM, and related LLM-in-education works generate lesson plans and instructional content from skills or knowledge graphs.

8. AI/ML/DL Model Identified by the Team:

- Model 1 – Student Proficiency Model

Type: Deep learning / ML model for knowledge tracing.

Input: Student quiz/assessment results (correctness, time, attempts).

Output: Skill-wise mastery scores for each student.

Possible approach: DKT-style RNN/transformer or a simpler gradient-boosted model over engineered features.

- Model 2 – Curriculum Design Model

Type: LLM-based generator (possibly fine-tuned or prompted).

Input: Course topic, level, duration, learning objectives, and industry skill set.

Output: Structured curriculum (modules, learning outcomes, topics, assessments, resources).

Possible approach: Prompted LLM with structured output schema, optionally fine-tuned on lesson-plan datasets.

- Model 3 – Industry Relevance Model

Type: NLP/LLM-based skill extraction and ranking model.

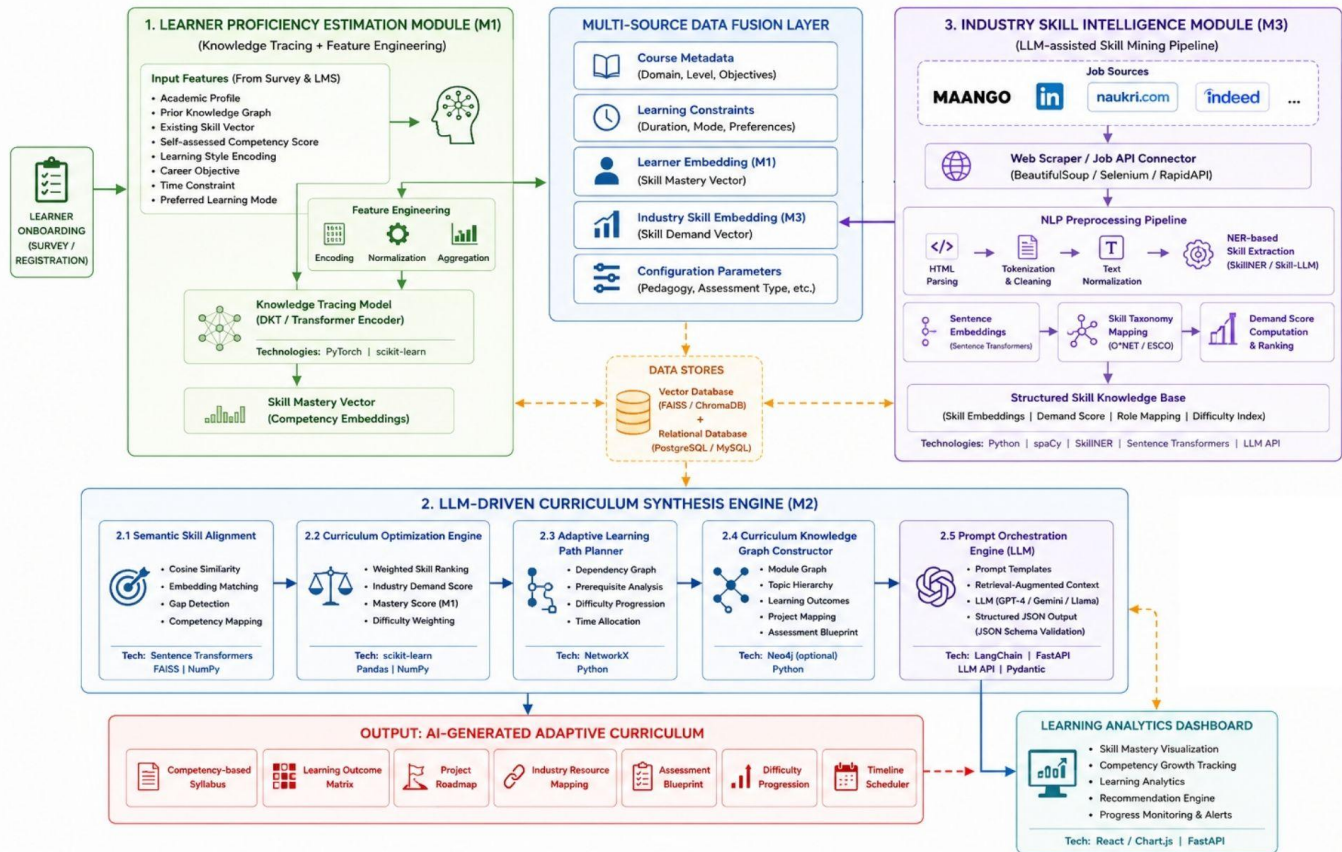
Input: Job descriptions from Mango, LinkedIn, etc.

Output: Structured list of required skills, tools, soft skills, and demand scores.

Possible approach: LLM or NER-based skill extractor (inspired by SKORE/Skill-LLM/SkillNer), followed by aggregation and ranking.

9. Model Architecture / Flow Diagram:

Automated curriculum design using LLMs



Signature of Supervisor

Signature of HOD